

Reconstruction of the Demographic History of South Asia from Himalayan archaeological sites using ancient DNA

Niraj Rai^{1,2}

¹Biological Sciences, DST-Birbal Sahni Institute of Palaeosciences, Lucknow, India,

²Biological Sciences, Academy of Scientific and Innovative Research, Ghaziabad, India



Introduction

- Burzahom, Kashmir shows distinct cultures of Early Neolithic, Late Neolithic, Megalithic and Early Historic Period.
- This site depicts various stages of development of agriculture and is one of the richest Neolithic cultures in South Asia.
- Seven high altitude archaeological sites in northern and central Himalayas, Lippa, Kanam, Sanglung, Tashigang, Kibber, Choskar and Suila ranging from ~2300 to 100 years old.

Methods

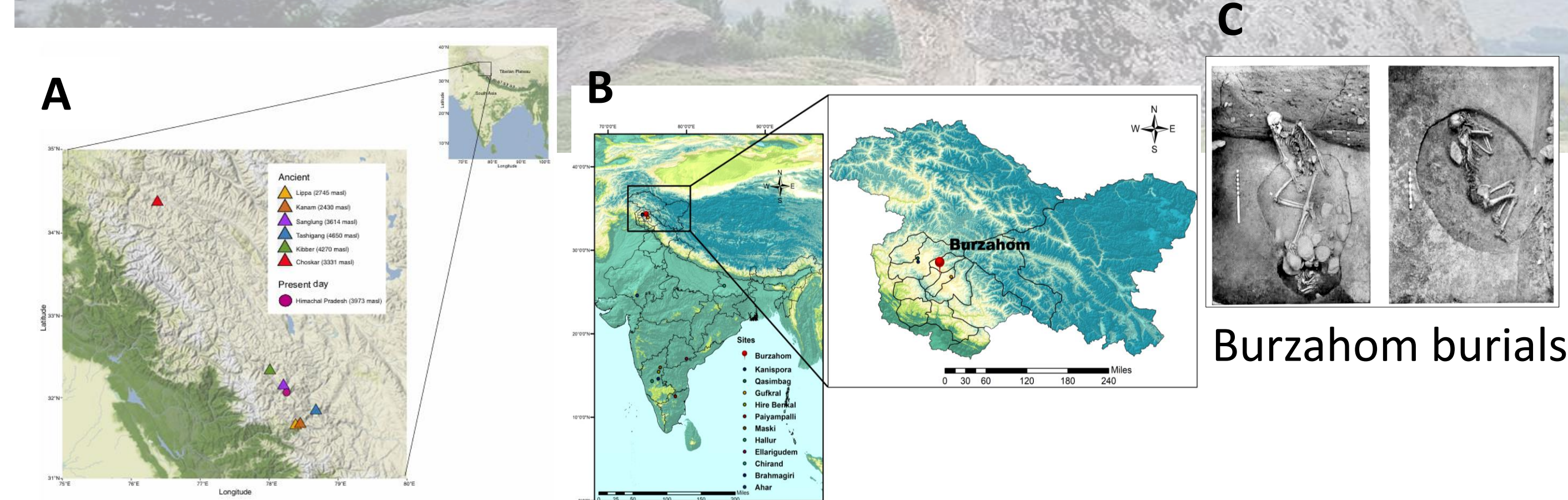
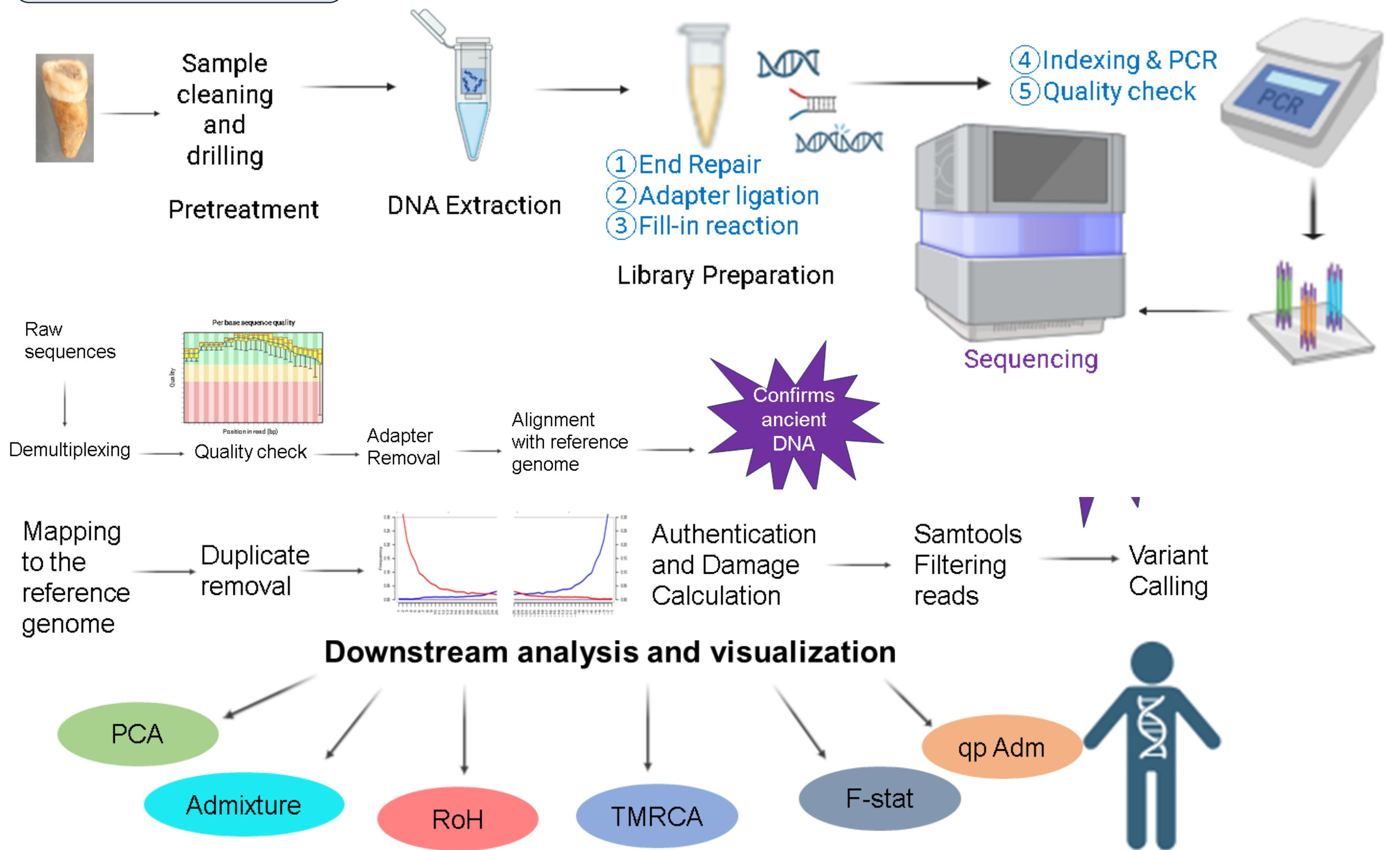


Fig. 1: A. Ancient Himalayan sites Lippa, Kanam, Sanglung, Tashigang, Kibber, Choskar B. Burzahom Neolithic Megalithic site from Kashmir

Fig. 2: Methods from drilling to data analysis

Results

Table 1: Summarized data from Burzahom

	BR_3	BR_7
Genetic Sex	Male	Male
Mt Haplogroup	M65a	U2b
% Endogenous	12.6%	12.4%
Mapped reads (mito)	47080	41069
Mean Coverage	201.96x	247.15x
MAP-authentic	0.9767	0.9761
Cal. ¹⁴ C age Mean	1927±32 BCE	563±20 CE

Principal Component Analysis (PCA)

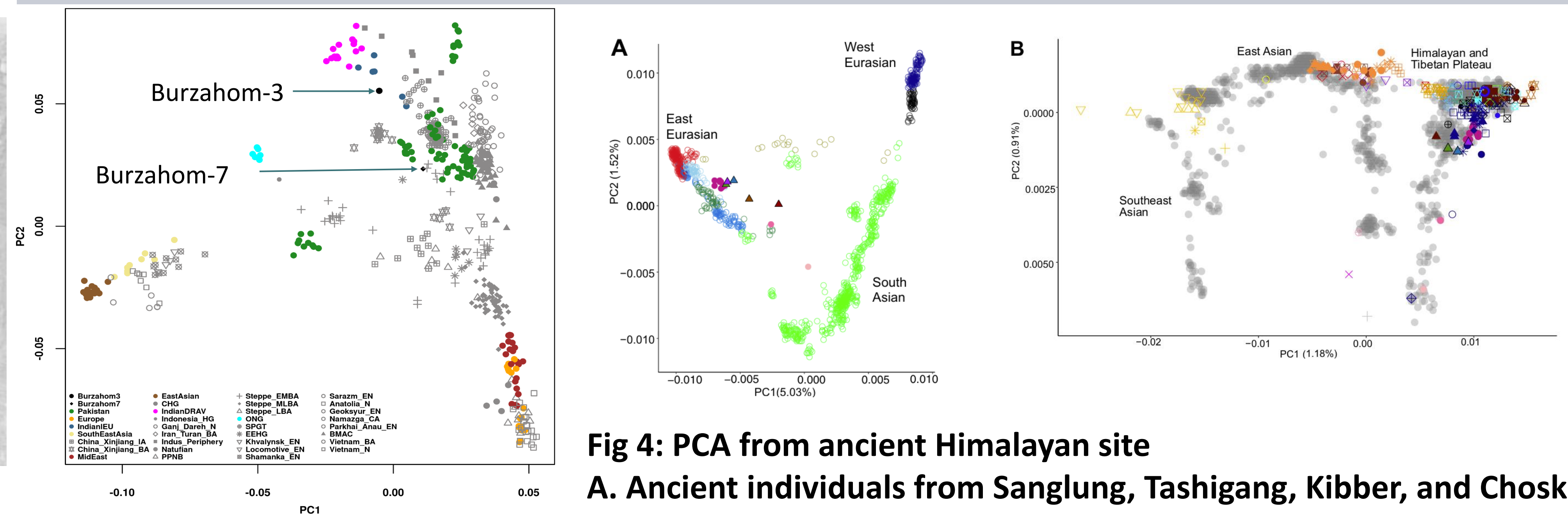


Fig 3: PCA of Burzahom individuals

Fig 4: PCA from ancient Himalayan site
A. Ancient individuals from Sanglung, Tashigang, Kibber, and Choskar were projected onto this PC space B. Eastern Eurasia-specific PCA

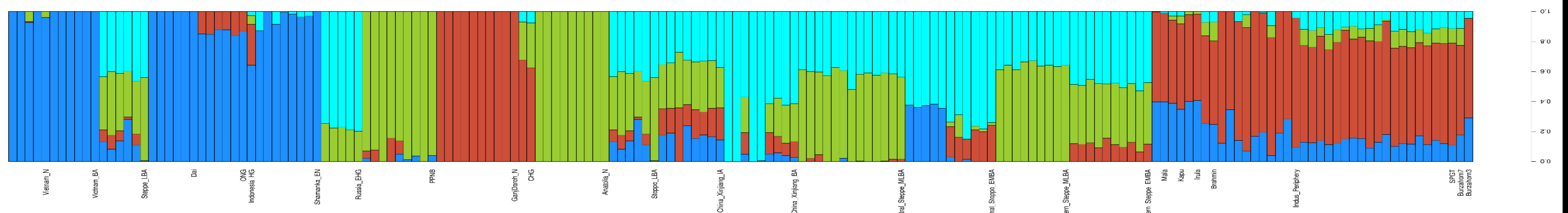


Fig. 5: Admixture Analysis of Burzahom individuals at K=4, colors represent different putative ancestral components

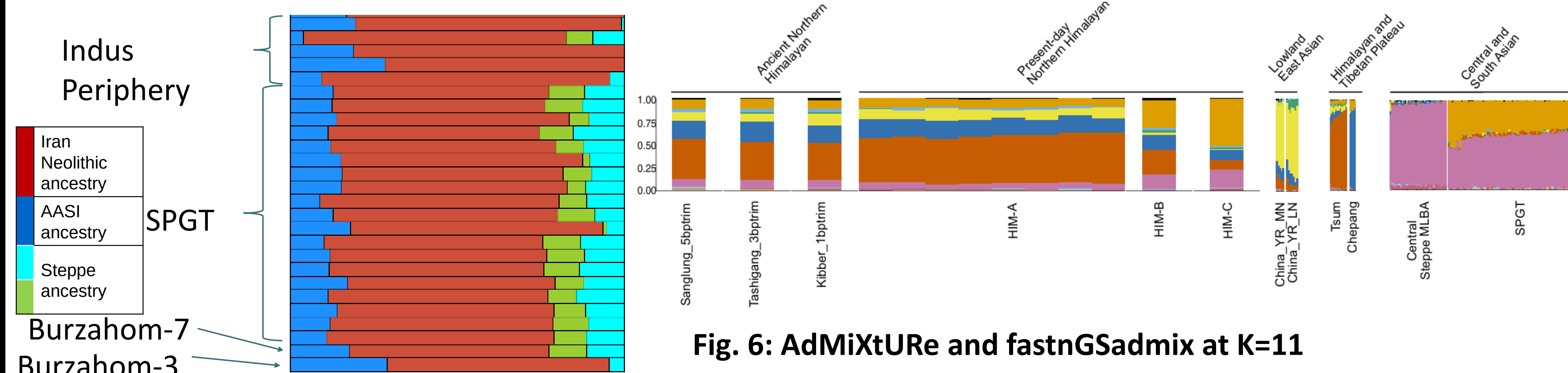


Fig. 6: AdMiXtUre and fastnGSadmix at K=11

Discussion

- Burz_3 (1927±32 BCE) is similar in ancestry to the Indus Periphery samples. Burz_7 (563±20 CE) shares ancestral components with Swat valley.
- Burz_7 is largely continuous with Burz_3, no major population turnover is detected.
- Ancient and present-day individuals from the northern Himalayas predominantly have Tibetan-related genetic ancestry, likely the source of high-altitude adaptive variants in these individuals, with substantial Steppe-related genetic ancestry that is observed in all individuals dating between ~1300 years and present day.

Fig. 7: Genetic sources and their proportions in the ancient northern himalayan individuals as inferred by qpAdm

Acknowledgements

- The director of DST-BSIP.
- Academy of scientific & innovative research
- Anthropological Survey of India